

Independent Model Validation Report

SME Probability-of-Default Model · SME term loans (synthetic example)

Prepared for	Demo Client
Engagement	DEMO-0005
Report date	11 November 2026
Version	1.0
Prepared by	ProQuant Analytics · info@proquantanalytics.com
Run ID	mv_20261111T142054Z_2fe5dd

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Executive summary

Demonstration report. Built entirely from synthetic data generated by ProQuant for illustration. The model, the lender and the loans are fictitious.

ProQuant Analytics independently validated the **SME Probability-of-Default Model** of Demo Client (SME term loans (synthetic example); purpose: Credit approval and risk-based pricing of SME loans up to 5m). The review covers the documentation, the data, replication of the model from its specification, performance on later data, challenger models, stability, sensitivity and performance by segment. Default = 90+ days past due within 12 months of origination.

Validation opinion: Not fit for use in its current form

2 high, 5 medium, 4 low severity finding(s) and 0 observation(s); 15 of 25 tests passed. Conclusions are drafted from the figures and reviewed by the analyst before release.

GINI, VALIDATION	KS, VALIDATION	OBSERVED / PREDICTED DEFAULTS
0.576	**0.441**	**1.25x**
development 0.566	AUC 0.788	validation sample

FINDINGS	DOCUMENTATION COVERAGE	GINI WITHOUT LEAKY INPUT
2 / 5 / 4	**74%**	**0.404**
high / medium / low	8 critical items	re-estimated, validation

Conclusions

1. Overall opinion: Not fit for use in its current form. 2 high, 5 medium and 4 low severity finding(s); 2 high-severity finding(s) must be remediated before the model is relied on.
2. Performance as the production model delivers it: Gini 0.576 and KS 0.441 on the 7,075 validation loans (development 0.566), but defaults are 1.25x the predicted level.
3. review_status is recorded after the lending decision and is in the model: removing it changes the picture (Gini 0.404 on validation instead of 0.576).
4. The production system differs from the documented specification for 2.64% of loans (mostly when current_ratio is missing).
5. The documentation overstates performance: validation Gini: reported 0.636, recalculated 0.576; validation KS: reported 0.481, recalculated 0.441.
6. The documented coefficient of leverage is 1.7x what the data support (z = 7.2); all other coefficients reproduce.
7. No challenger beats the model on validation (best advantage -0.176): the benchmark logistic and gradient boosting reach a Gini of about 0.37 to 0.40; the model's higher Gini is explained by the leaky input.
8. Stability: PD PSI 0.075 in 2023Q2; largest input shift leverage (CSI 0.13). Weakest segment: business_age = under 2 (Gini 74% of the overall).
9. Without review_status, the documented specification re-estimated on the development sample reaches Gini 0.404 and KS 0.321 on the validation sample: above the minimum standard (Gini 0.35), so the model can be used once the input is removed and the findings are remediated.

Findings

ID	Severity	Area	Finding	Due
F-01	High	Data	An input is recorded after the lending decision (leakage)	2026-12-11
F-02	High	Replication	Reported discrimination is overstated	2026-12-11
F-03	Medium	Replication	Documented coefficient not supported by the data	2027-02-09
F-04	Medium	Replication	Production PD differs from the documented specification	2027-02-09
F-05	Medium	Replication	Reported calibration does not match the data	2027-02-09
F-06	Medium	Performance	Model under- or over-predicts defaults	2027-02-09
F-07	Medium	Documentation	Critical documentation items missing or partial	2027-02-09
F-08	Low	Stability	An input's distribution has shifted	2027-05-10
F-09	Low	Segments	The model is weak for a segment	2027-05-10
F-10	Low	Documentation	Documentation is incomplete	2027-05-10
F-11	Low	Governance	Model used beyond its documented scope	2027-05-10

Source: tables/findings.csv. High: do not rely on the model until fixed; Medium: fix within 90 days with a compensating control; Low: improvement within 180 days.

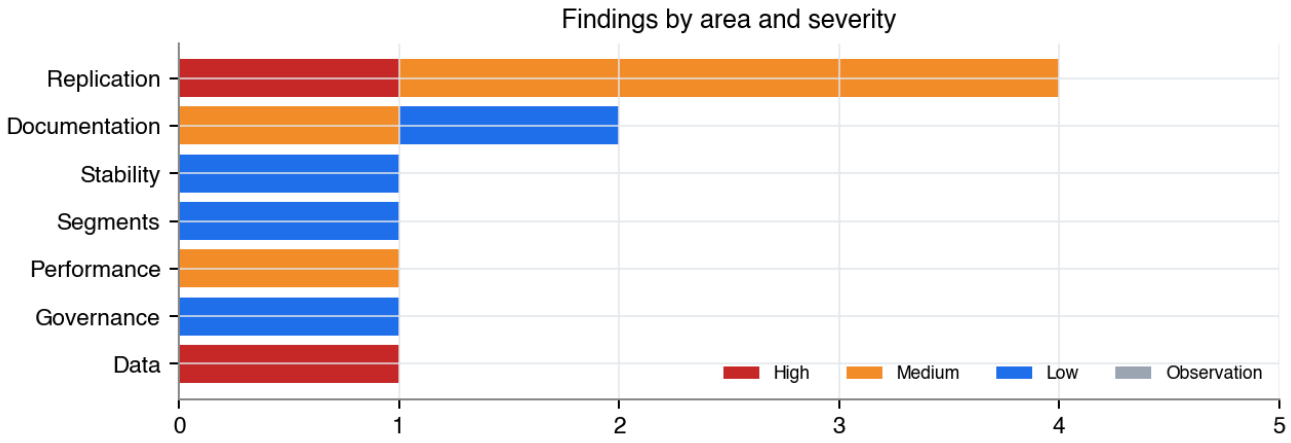


Figure 1. Findings by area and severity. Source: tables/findings.csv

Methodology

This report sets out the opinion and the evidence behind it. The steps, formulas, thresholds and references are in the methodology document.

Results

1. All tests

Area	Test	Result	Threshold	Status
Documentation	Documentation coverage of the checklist	74%	>= 0.85 pass, >= 0.7 warn	WARN
Documentation	Critical documentation items are all met	3 missing, 0 partial	none missing or partial	FAIL
Data	No input is recorded after the decision (leakage)	1 input(s)	none	FAIL
Data	Largest single-input information value	0.109	<= 0.5 pass, <= 1 warn	PASS
Data	Defaults in the development sample	887	>= 500 pass, >= 300 warn	PASS
Data	Largest missing rate among model inputs	2.7%	<= 0.05 pass, <= 0.15 warn	PASS
Data	Representativeness: PD PSI development vs validation	0.080	<= 0.1 pass, <= 0.25 warn	PASS
Data	Data-quality checks on the inputs	passed	no error-level issue	PASS
Replication	Production PD equals the PD recomputed from the specification (rows differing)	2.64%	<= 0.001 pass, <= 0.05 warn	WARN
Replication	Reported Gini / KS equal recalculated (largest gap)	0.060	<= 0.02 pass, <= 0.05 warn	FAIL
Replication	Reported calibration equals recalculated (gap in obs / pred)	0.245	<= 0.05 pass, <= 0.15 warn	FAIL
Replication	Documented coefficients supported by the data (largest |z|)	7.2	<= 3 pass, <= 5 warn	FAIL
Performance	Gini on the validation sample	0.576	>= 0.35 pass, >= 0.25 warn	PASS
Performance	KS on the validation sample	0.441	>= 0.25 pass, >= 0.15 warn	PASS
Performance	Gini drop from development to validation	-0.009	<= 0.05 pass, <= 0.1 warn	PASS
Performance	Calibration: observed / predicted defaults, validation	1.25x	|ratio - 1| <= 0.2 pass, <= 0.35 warn	WARN
Challengers	Best challenger's Gini advantage over the model	-0.176	<= 0.03 pass, <= 0.06 warn	PASS
Stability	PD PSI, latest quarter	0.075	<= 0.1 pass, <= 0.25 warn	PASS
Stability	Largest input CSI, latest quarter	0.130	<= 0.1 pass, <= 0.25 warn	WARN
Stability	Rank-ordering: worst quarter's Gini below the overall	0.033	<= 0.08 pass, <= 0.15 warn	PASS
Stability	Width of the 95% bootstrap interval of the validation Gini	0.016	<= 0.06 pass, <= 0.12 warn	PASS
Sensitivity	Largest Gini lost by dropping one term	0.061	<= 0.08 pass, <= 0.15 warn	PASS
Sensitivity	Largest change of mean PD for a 10% shock to one input	10%	<= 0.25 pass, <= 0.5 warn	PASS
Segments	Weakest segment's Gini as a share of the overall	74%	>= 0.75 pass, >= 0.55 warn	WARN
Segments	Worst segment calibration relative to the overall (|ratio / overall ratio - 1|)	0.17	<= 0.25 pass, <= 0.4 warn	PASS

Source: tables/validation_tests.csv

2. Documentation

Test	Result	Threshold	Status
Documentation coverage of the checklist	74%	>= 0.85 pass, >= 0.7 warn	WARN
Critical documentation items are all met	3 missing, 0 partial	none missing or partial	FAIL

Source: tables/validation_tests.csv

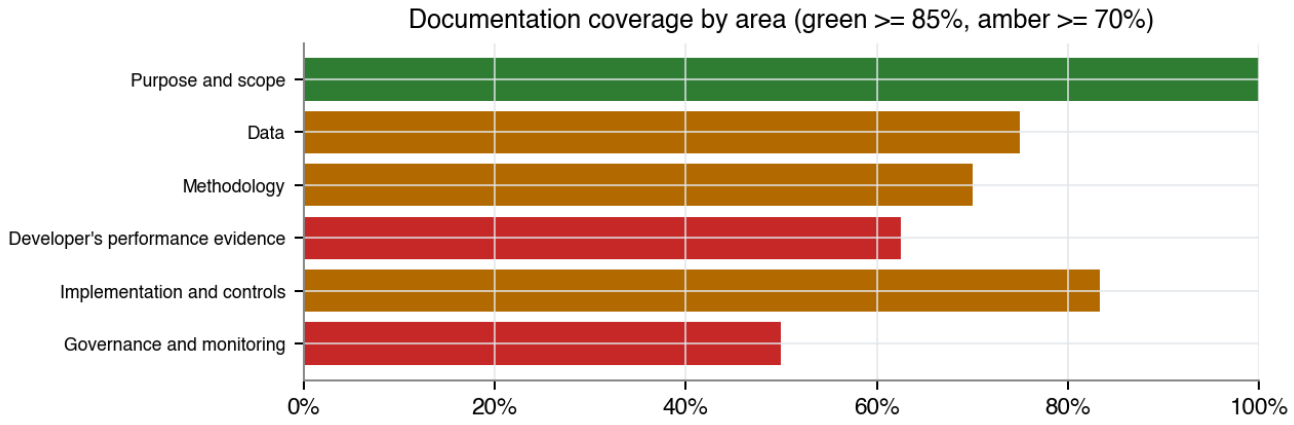


Figure 2. Coverage by area. Source: tables/documentation_by_area.csv

ID	Area	Item	Critical	Rating	Evidence
D01	Purpose and scope	Intended use, users and decisions the model supports are stated	yes	PASS	Development report s.1.2
D02	Purpose and scope	Portfolio and population covered, and exclusions, are defined	yes	PASS	Development report s.1.3
D03	Purpose and scope	Definition of default and the outcome window are documented	yes	PASS	Development report s.2.1: 90+ days past due within 12 months
D04	Purpose and scope	Regulatory and policy requirements the model must meet are identified		PASS	Development report s.1.4 maps the model to the credit policy
D05	Data	Data sources and extraction dates are documented		PASS	Development report s.2.2
D06	Data	Time of availability of every input (at decision / later) is documented	yes	FAIL	No field dictionary; review_status timing not stated
D07	Data	Data cleaning, exclusions and treatment of missing values are described		PASS	Development report s.2.4
D08	Data	Representativeness of the development sample is assessed		PASS	Development report s.2.5: default rate and mix by year and sector
D09	Methodology	Modelling approach and its rationale are described		PASS	Development report s.3.1
D10	Methodology	Variable selection and transformations are documented and reproducible	yes	PASS	model_spec.csv and report s.3.3
D11	Methodology	Coefficients, signs and their business rationale are documented		PASS	Report table 6
D12	Methodology	Assumptions and limitations of the model are stated	yes	FAIL	No section on assumptions or limitations
D13	Methodology	Alternatives considered (other methods, variables) are described		WARN	A gradient-boosting test is mentioned without results
D14	Developer's performance evidence	Discrimination (Gini / KS) on development and hold-out data is reported		PASS	Report table 4 and back-test memo
D15	Developer's performance evidence	Calibration of the PD to observed default rates is reported		PASS	Back-test memo table 3
D16	Developer's performance evidence	Out-of-time performance and stability are reported		WARN	One out-of-time year, no breakdown by quarter
D17	Developer's performance evidence	Performance by segment (sector, size, age of business) is reported		FAIL	Not reported
D18	Implementation and controls	The specification is complete enough to re-implement the model	yes	PASS	model_spec.csv; fill values documented
D19	Implementation and controls	Production implementation is tested against the specification		WARN	A test of 50 cases is mentioned without results
D20	Implementation and controls	Input controls and handling of missing / invalid data in production are described		PASS	Operations manual s.4
D21	Governance and monitoring	Model owner, developer and approver are identified		PASS	Model inventory entry
D22	Governance and monitoring	A monitoring plan with metrics and thresholds exists	yes	FAIL	No monitoring plan; the annual back-test memo is the only monitoring
D23	Governance and monitoring	Change control and version history are documented		WARN	Version 1.2 noted; no change log

ID	Area	Item	Critical	Rating	Evidence
D24	Governance and monitoring	Date and scope of the last validation and open issues are recorded		N/A	First validation of this model

Source: tables/documentation_review.csv

3. Data

Test	Result	Threshold	Status
No input is recorded after the decision (leakage)	1 input(s)	none	FAIL
Largest single-input information value	0.109	<= 0.5 pass, <= 1 warn	PASS
Defaults in the development sample	887	>= 500 pass, >= 300 warn	PASS
Largest missing rate among model inputs	2.7%	<= 0.05 pass, <= 0.15 warn	PASS
Representativeness: PD PSI development vs validation	0.080	<= 0.1 pass, <= 0.25 warn	PASS
Data-quality checks on the inputs	passed	no error-level issue	PASS

Source: tables/validation_tests.csv

Sample	Loans	First	Last	Defaults	Default rate
development	21,451	2019-01-01	2021-12-31	887	4.14%
monitoring	3,525	2023-01-01	2023-06-30	274	7.77%
validation	7,075	2022-01-01	2022-12-31	499	7.05%

Source: tables/samples.csv

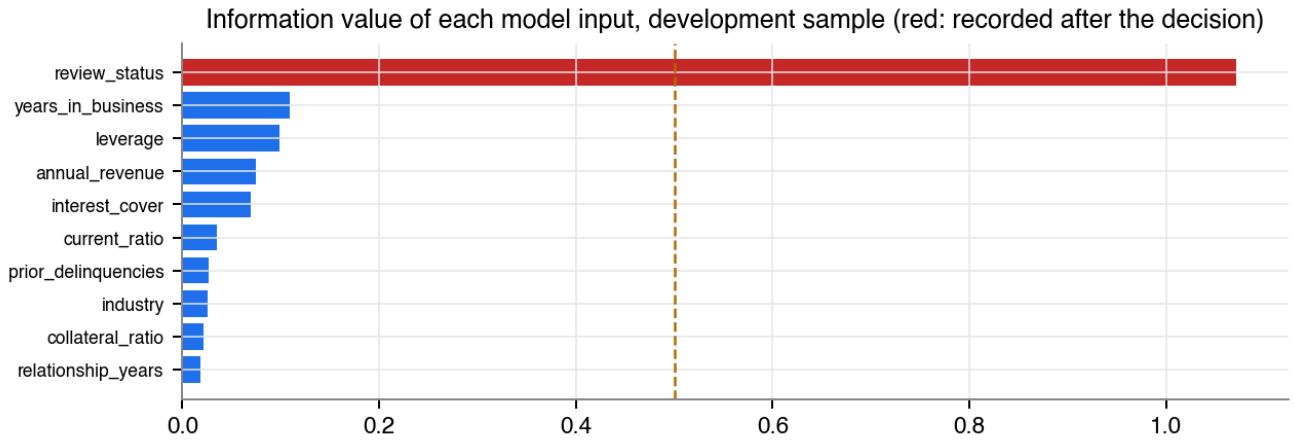


Figure 3. Information value of the model inputs. Source: tables/input_screen.csv

Input	Known at	Missing	IV	Univariate Gini	After decision
review_status	post_decision	0.0%	1.071	0.378	True
years_in_business	application	0.0%	0.109	0.165	False
leverage	application	0.0%	0.098	0.160	False
annual_revenue	application	0.0%	0.075	0.147	False
interest_cover	application	2.1%	0.069	0.141	False
current_ratio	application	2.7%	0.035	0.077	False
prior_delinquencies	application	0.0%	0.027	0.071	False
industry	application	0.0%	0.025	0.089	False
collateral_ratio	application	0.0%	0.022	0.066	False
relationship_years	application	1.4%	0.019	0.057	False

Source: tables/input_screen.csv

review_status is recorded after the loan is granted (stated by the lender in the field timing) and is used by the model. A decision model can only use what is known when the decision is taken.

4. Replication

Test	Result	Threshold	Status
Production PD equals the PD recomputed from the specification (rows differing)	2.64%	<= 0.001 pass, <= 0.05 warn	WARN
Reported Gini / KS equal recalculated (largest gap)	0.060	<= 0.02 pass, <= 0.05 warn	FAIL
Reported calibration equals recalculated (gap in obs / pred)	0.245	<= 0.05 pass, <= 0.15 warn	FAIL
Documented coefficients supported by the data (largest |z|)	7.2	<= 3 pass, <= 5 warn	FAIL

Source: tables/validation_tests.csv

4.1 Implementation: production PD vs the specification

845 of 32,051 loans (2.64%) have a production PD that differs from the PD recomputed from the documentation by more than 0.0001 (largest difference 0.0230; correlation 0.99998).

Missing input	Rows missing	Share of rows	Rows differing when missing	Rows differing otherwise	Share of all differences
current_ratio	837	2.6%	100.0%	0.03%	99.1%
interest_cover	672	2.1%	3.3%	2.62%	2.6%
relationship_years	465	1.5%	2.4%	2.64%	1.3%
years_in_business	6	0.0%	100.0%	2.62%	0.7%
annual_revenue	2	0.0%	100.0%	2.63%	0.2%

Source: tables/implementation_diagnosis.csv

4.2 Re-estimation of the coefficients

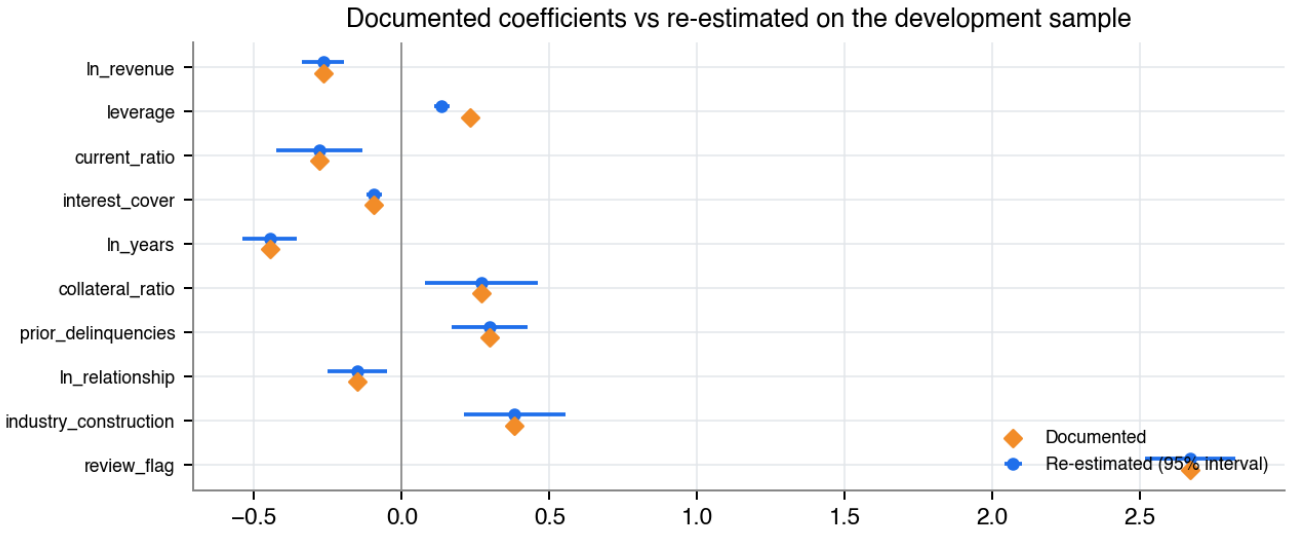


Figure 4. Documented vs re-estimated coefficients. Source: tables/coefficient_replication.csv

Term	Documented	Re-estimated	Std error	Difference	z	p-value	Documented / re-est.	Same sign
intercept	0.579	1.008	0.538	0.429	0.8	0.425	0.57	True
ln_revenue	-0.267	-0.267	0.037	-0.000	-0.0	1.000	1.00	True
leverage	0.231	0.136	0.013	-0.095	-7.2	0.000	1.70	True
current_ratio	-0.279	-0.279	0.074	0.000	0.0	0.999	1.00	True
interest_cover	-0.094	-0.094	0.013	0.000	0.0	0.999	1.00	True
ln_years	-0.447	-0.447	0.047	0.000	0.0	0.997	1.00	True
collateral_ratio	0.269	0.269	0.097	0.000	0.0	0.999	1.00	True
prior_delinquencies	0.297	0.297	0.065	0.000	0.0	0.999	1.00	True
ln_relationship	-0.151	-0.151	0.051	-0.000	-0.0	0.999	1.00	True
industry_construction	0.382	0.382	0.088	0.000	0.0	0.999	1.00	True
review_flag	2.668	2.668	0.078	0.000	0.0	0.999	1.00	True

Source: tables/coefficient_replication.csv

4.3 Reported vs recalculated

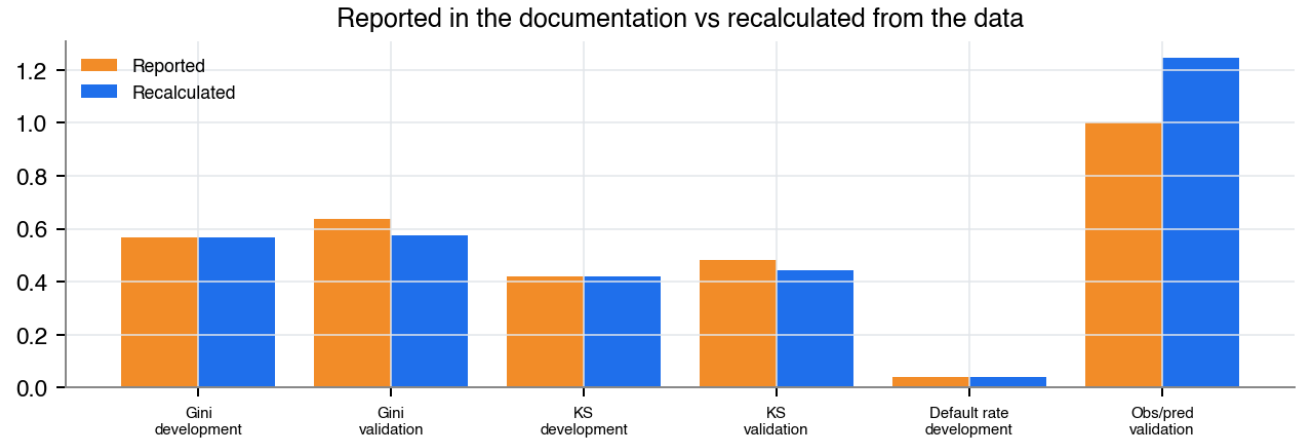


Figure 5. Source: tables/reported_vs_recalculated.csv

Metric	Sample	Reported	Recalculated	Difference	Source	Status
Gini	development	0.566	0.566	-0.000	Model development report, table 4	PASS
Gini	validation	0.636	0.576	+0.060	Annual back-test memo, table 1	FAIL
KS	development	0.420	0.420	-0.000	Model development report, table 4	PASS
KS	validation	0.481	0.441	+0.040	Annual back-test memo, table 1	WARN
Default rate	development	0.041	0.041	+0.000	Model development report, table 2	PASS
Defaults	development	887.000	887.000	+0.000	Model development report, table 2	PASS
Observed / predicted default rate	validation	1.000	1.245	-0.245	Annual back-test memo, table 3	FAIL

Source: tables/reported_vs_recalculated.csv

5. Performance

Test	Result	Threshold	Status
Gini on the validation sample	0.576	>= 0.35 pass, >= 0.25 warn	PASS
KS on the validation sample	0.441	>= 0.25 pass, >= 0.15 warn	PASS
Gini drop from development to validation	-0.009	<= 0.05 pass, <= 0.1 warn	PASS
Calibration: observed / predicted defaults, validation	1.25x	|ratio - 1| <= 0.2 pass, <= 0.35 warn	WARN

Source: tables/validation_tests.csv

Sample	Loans	Defaults	Default rate	Mean PD	Obs / pred	AUC	Gini	Gini 95% interval	KS
development	21,451	887	4.14%	4.14%	1.00	0.783	0.566	0.529 to 0.601	0.420
validation	7,075	499	7.05%	5.66%	1.25	0.788	0.576	0.527 to 0.613	0.441
monitoring	3,525	274	7.77%	5.81%	1.34	0.800	0.601	0.540 to 0.654	0.457

Source: tables/performance_summary.csv

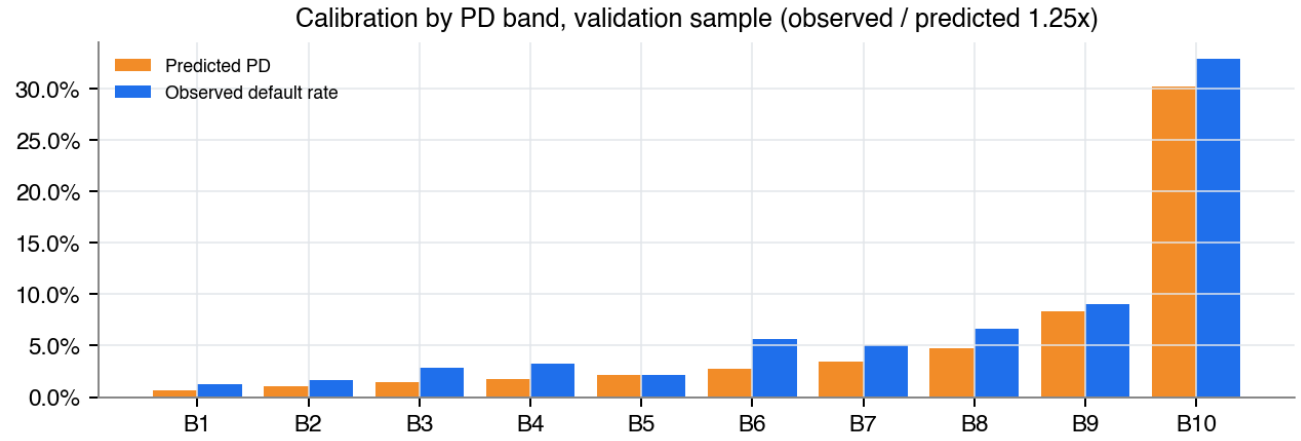


Figure 6. Calibration. Source: tables/calibration_validation.csv

Band	Loans	Defaults	Predicted PD	Observed	Obs / pred	p-value
1	708	9	0.68%	1.27%	1.86	0.12
2	707	12	1.10%	1.70%	1.55	0.19
3	708	20	1.44%	2.82%	1.96	0.01
4	707	23	1.79%	3.25%	1.82	0.01
5	708	15	2.19%	2.12%	0.97	1.03
6	707	40	2.70%	5.66%	2.10	0.00
7	707	36	3.43%	5.09%	1.49	0.03
8	708	47	4.76%	6.64%	1.40	0.03
9	707	64	8.30%	9.05%	1.09	0.51
10	708	233	30.23%	32.91%	1.09	0.13

Source: tables/calibration_validation.csv

6. Challengers

Test	Result	Threshold	Status
Best challenger's Gini advantage over the model	-0.176	<= 0.03 pass, <= 0.06 warn	PASS

Source: tables/validation_tests.csv

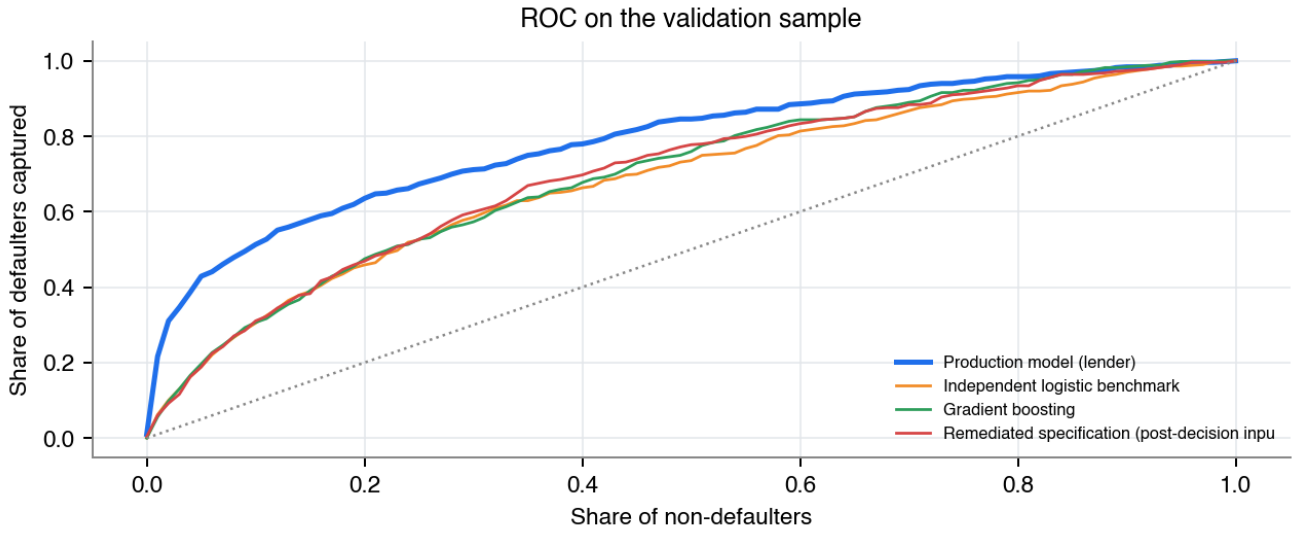


Figure 7. ROC curves. Source: tables/roc_validation.csv

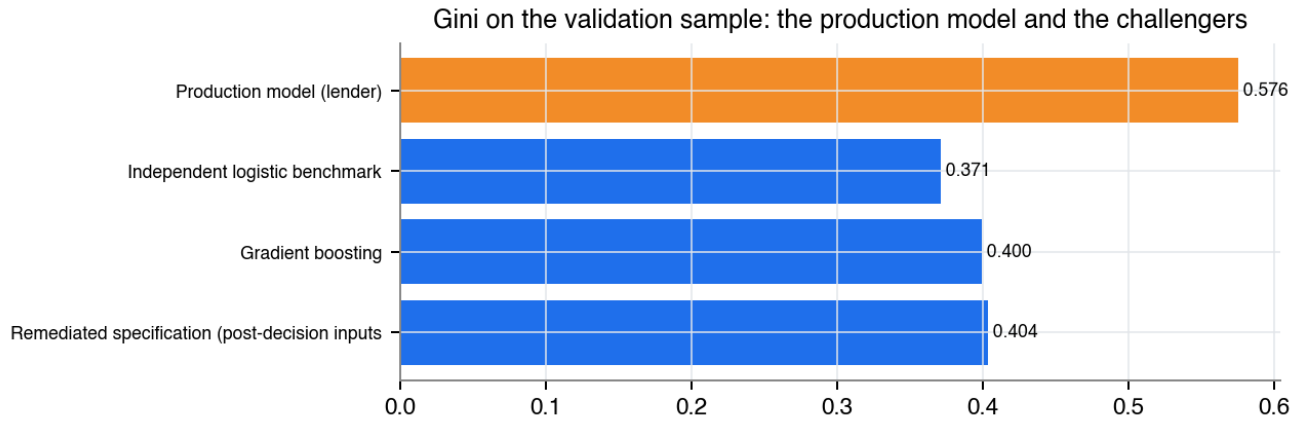


Figure 8. Gini by model. Source: tables/challenger_comparison.csv

Model	Loans	Defaults	Gini	KS	AUC	Brier	Mean PD	Obs / pred	Sample
Production model (lender)	21,451	887	0.566	0.420	0.783	0.0345	4.14%	1.00	development
Independent logistic benchmark	21,451	887	0.351	0.250	0.675	0.0389	4.14%	1.00	development
Gradient boosting	21,451	887	0.533	0.389	0.767	0.0378	4.12%	1.00	development
Remediated specification (post-decision inputs removed)	21,451	887	0.379	0.285	0.689	0.0389	4.14%	1.00	development
Production model (lender)	7,075	499	0.576	0.441	0.788	0.0539	5.66%	1.25	validation
Independent logistic benchmark	7,075	499	0.371	0.292	0.686	0.0637	4.91%	1.44	validation
Gradient boosting	7,075	499	0.400	0.288	0.700	0.0637	4.73%	1.49	validation
Remediated specification (post-decision inputs removed)	7,075	499	0.404	0.321	0.702	0.0634	4.90%	1.44	validation

Source: tables/challenger_comparison.csv

If the finding is remediated

Without review_status, the documented specification re-estimated on the development sample reaches Gini 0.404 and KS 0.321 on the validation sample: above the minimum standard (Gini 0.35), so the model can be used once the input is removed and the findings are remediated.

7. Stability

Test	Result	Threshold	Status
PD PSI, latest quarter	0.075	<= 0.1 pass, <= 0.25 warn	PASS
Largest input CSI, latest quarter	0.130	<= 0.1 pass, <= 0.25 warn	WARN
Rank-ordering: worst quarter's Gini below the overall	0.033	<= 0.08 pass, <= 0.15 warn	PASS
Width of the 95% bootstrap interval of the validation Gini	0.016	<= 0.06 pass, <= 0.12 warn	PASS

Source: tables/validation_tests.csv

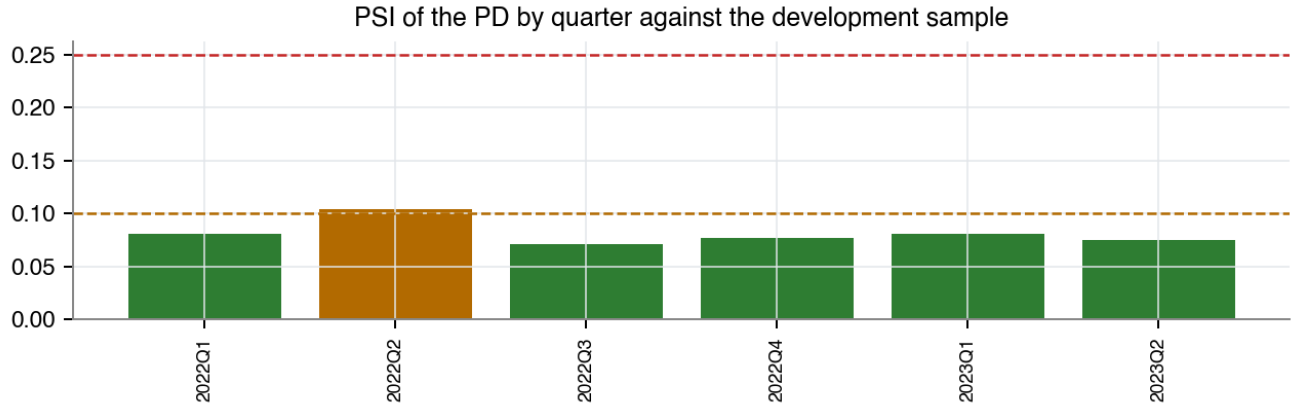


Figure 9. PD PSI by quarter. Source: tables/psi_pd.csv

Quarter	Loans	Mean PD	PSI	Status
2022Q1	1,832	5.55%	0.081	PASS
2022Q2	1,673	5.86%	0.104	WARN
2022Q3	1,825	5.49%	0.071	PASS
2022Q4	1,745	5.77%	0.077	PASS
2023Q1	1,813	5.85%	0.081	PASS
2023Q2	1,712	5.75%	0.075	PASS

Source: tables/psi_pd.csv

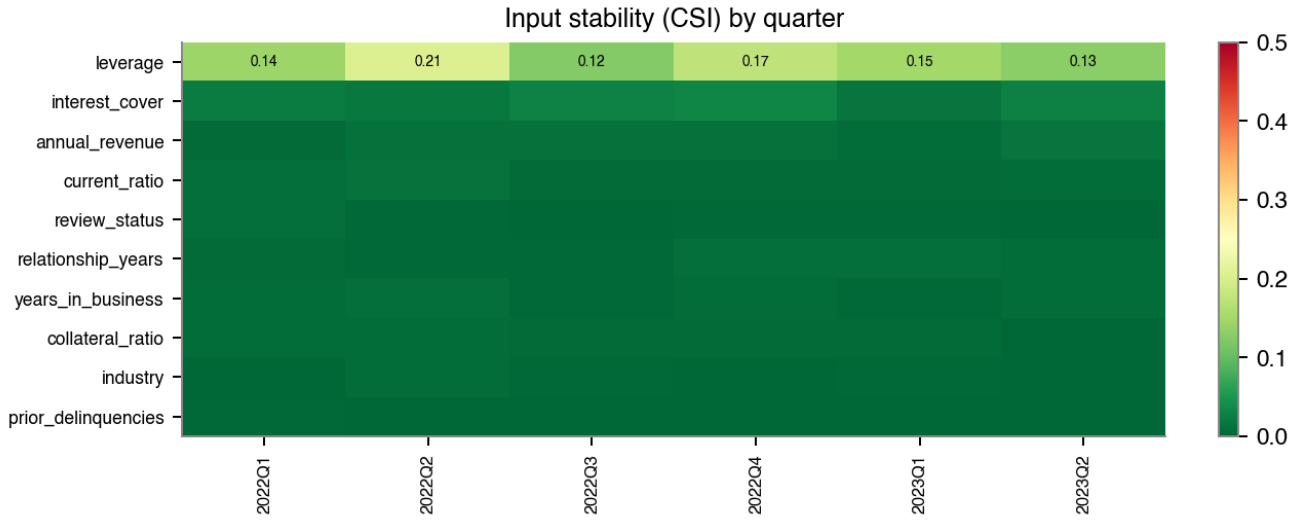


Figure 10. Input CSI by quarter. Source: tables/csi.csv

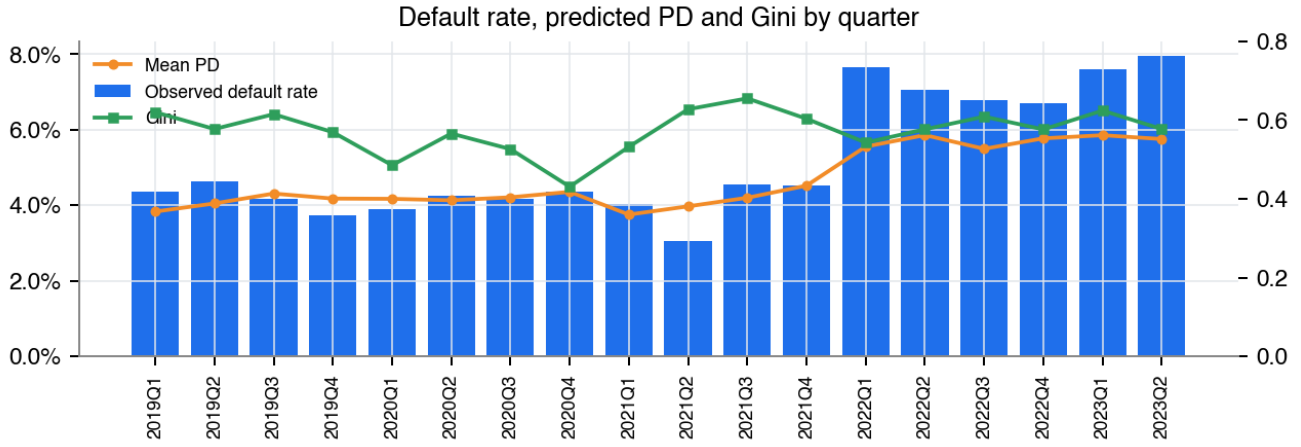


Figure 11. Performance by quarter. Source: tables/performance_by_period.csv

Quarter	Loans	Defaults	Default rate	Mean PD	Obs / pred	Gini
2019Q1	1,766	77	4.36%	3.83%	1.14	0.620
2019Q2	1,751	81	4.63%	4.05%	1.14	0.577
2019Q3	1,756	73	4.16%	4.31%	0.96	0.615
2019Q4	1,799	67	3.72%	4.18%	0.89	0.569
2020Q1	1,748	68	3.89%	4.17%	0.93	0.485
2020Q2	1,855	79	4.26%	4.13%	1.03	0.566
2020Q3	1,900	79	4.16%	4.21%	0.99	0.526
2020Q4	1,768	77	4.36%	4.35%	1.00	0.431
2021Q1	1,760	71	4.03%	3.75%	1.07	0.532
2021Q2	1,835	56	3.05%	3.97%	0.77	0.627
2021Q3	1,718	78	4.54%	4.20%	1.08	0.655
2021Q4	1,795	81	4.51%	4.52%	1.00	0.603
2022Q1	1,832	140	7.64%	5.55%	1.38	0.543
2022Q2	1,673	118	7.05%	5.86%	1.20	0.577
2022Q3	1,825	124	6.79%	5.49%	1.24	0.609
2022Q4	1,745	117	6.70%	5.77%	1.16	0.576
2023Q1	1,813	138	7.61%	5.85%	1.30	0.624
2023Q2	1,712	136	7.94%	5.75%	1.38	0.577

Source: tables/performance_by_period.csv

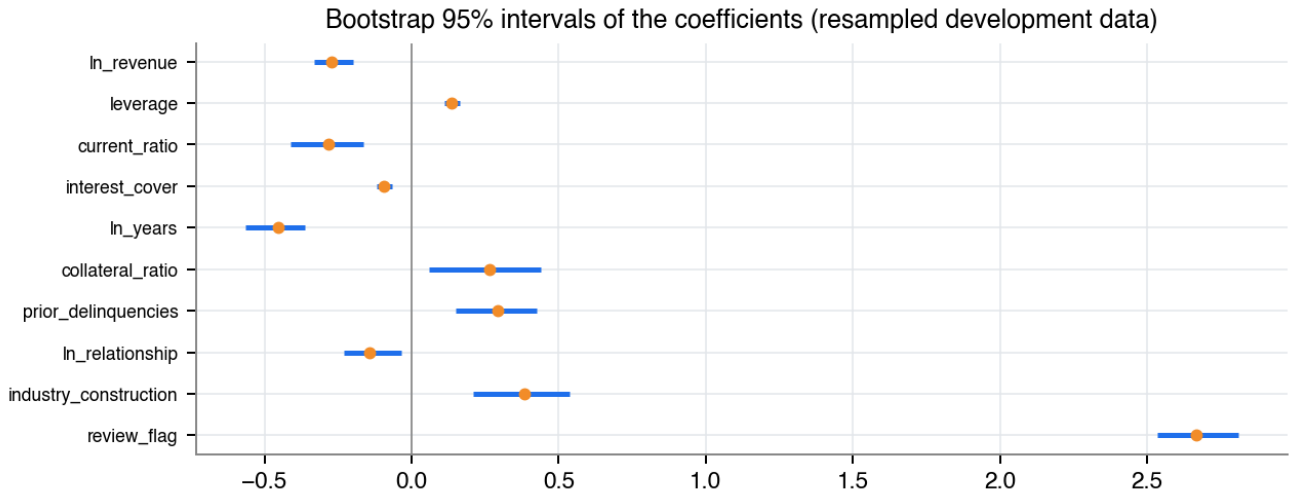


Figure 12. Bootstrap intervals (100 resamples; validation Gini 0.561 to 0.576). Source: tables/bootstrap_coefficients.csv

8. Sensitivity

Test	Result	Threshold	Status
Largest Gini lost by dropping one term	0.061	<= 0.08 pass, <= 0.15 warn	PASS
Largest change of mean PD for a 10% shock to one input	10%	<= 0.25 pass, <= 0.5 warn	PASS

Source: tables/validation_tests.csv

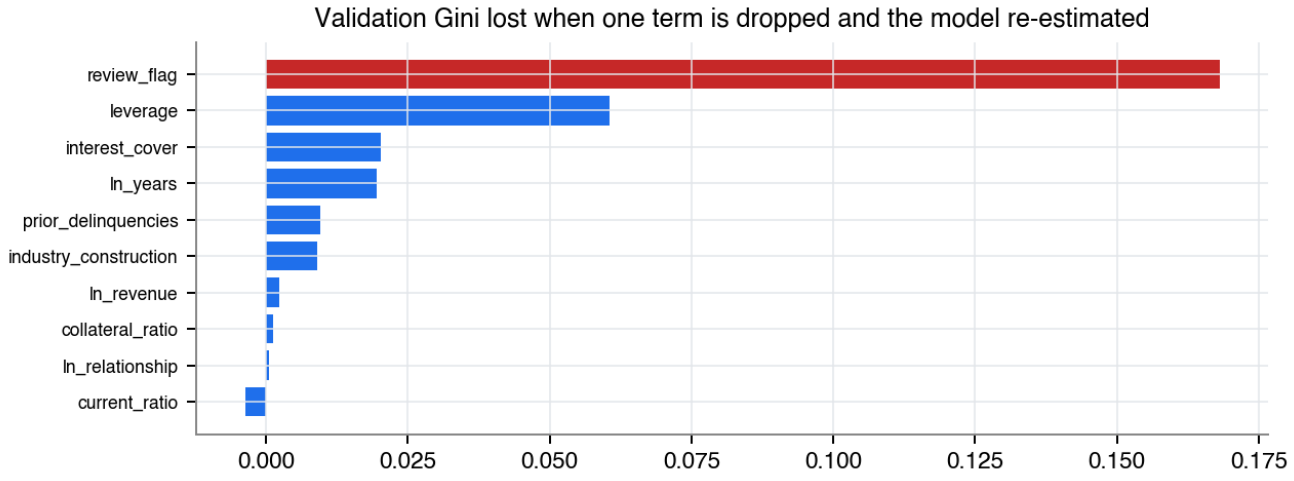


Figure 13. Gini lost per dropped term. Source: tables/variable_drop.csv

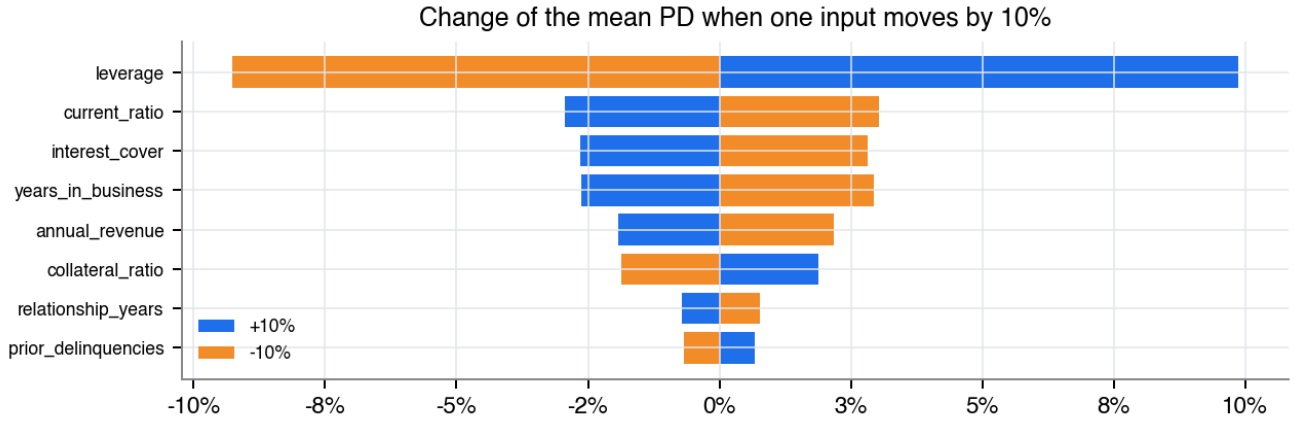


Figure 14. Input shocks. Source: tables/input_shocks.csv

Stress scenarios

Scenario	Moves	Mean PD	Change vs base	Loans with PD > 10%
Downturn: revenue -20%, leverage +30%	annual_revenue x0.8, leverage x1.3	7.73%	36.6%	18.7%
Rates up: interest cover -25%	interest_cover x0.75	6.08%	7.4%	13.1%
Severe: revenue -30%, leverage +50%, interest cover -40%	annual_revenue x0.7, leverage x1.5, interest_cover x0.6	10.31%	82.2%	26.8%

Source: tables/stress_scenarios.csv

9. Segments

Test	Result	Threshold	Status
Weakest segment's Gini as a share of the overall	74%	>= 0.75 pass, >= 0.55 warn	WARN
Worst segment calibration relative to the overall (|ratio / overall ratio - 1|)	0.17	<= 0.25 pass, <= 0.4 warn	PASS

Source: tables/validation_tests.csv



Figure 15. Gini by segment. Source: tables/segments.csv

Segment	Group	Loans	Defaults	Default rate	Mean PD	Obs / pred	Gini
business_age	10 and over	2,708	157	5.80%	4.07%	1.42	0.572
business_age	2 to 5	905	66	7.29%	7.09%	1.03	0.629
business_age	5 to 10	2,602	178	6.84%	5.78%	1.18	0.598
business_age	under 2	860	98	11.40%	8.83%	1.29	0.425
size	1e+06 to 5e+06	2,894	185	6.39%	4.75%	1.35	0.548
size	250000 to 1e+06	3,208	223	6.95%	6.18%	1.12	0.575
size	5e+06 and over	241	16	6.64%	3.37%	1.97	n/a
size	under 250000	732	75	10.25%	7.75%	1.32	0.648
industry	Agriculture	366	24	6.56%	5.34%	1.23	n/a
industry	Construction	1,211	118	9.74%	7.55%	1.29	0.544
industry	Manufacturing	1,409	98	6.96%	5.59%	1.24	0.620
industry	Retail	1,578	101	6.40%	5.57%	1.15	0.625
industry	Services	1,778	111	6.24%	5.10%	1.22	0.480
industry	Transport	733	47	6.41%	4.43%	1.45	0.557
region	Central	1,063	95	8.94%	6.14%	1.45	0.542
region	East	1,405	84	5.98%	5.34%	1.12	0.457
region	North	1,831	136	7.43%	5.60%	1.33	0.597
region	South	1,365	95	6.96%	5.72%	1.22	0.579
region	West	1,411	89	6.31%	5.66%	1.12	0.684

Source: tables/segments.csv

10. Detailed findings

F-01 High An input is recorded after the lending decision (leakage)

review_status is recorded after the loan is granted and reflects its performance, so it cannot be known when the model is used. It carries 0.168 of the model's validation Gini (re-estimated without it the Gini is 0.404 against 0.576 reported by the production model), so the model's apparent accuracy is overstated.

Recommendation. Remove the input and re-estimate the model; re-run the validation. Introduce a field dictionary that states when each input is known.

Area: Data · Evidence: tables/input_screen.csv · Owner: Model owner · Due: 2026-12-11 · Status: Open

F-02 High Reported discrimination is overstated

validation Gini: reported 0.636, recalculated 0.576; validation KS: reported 0.481, recalculated 0.441

Recommendation. Correct the documentation and the back-test memo; re-compute the metrics on the out-of-time sample with the validation team.

Area: Replication · Evidence: tables/reported_vs_recalculated.csv · Owner: Model owner · Due: 2026-12-11 · Status: Open

F-03 Medium Documented coefficient not supported by the data

Re-estimating the documented specification on the development sample gives leverage = 0.136 (documented 0.231, 1.7x; z = 7.2). The other coefficients reproduce.

Recommendation. Explain the difference (different sample, filter or software) or correct the documented value; assess the effect on the PDs.

Area: Replication · Evidence: tables/coefficient_replication.csv · Owner: Model owner · Due: 2027-02-09 · Status: Open

F-04 Medium Production PD differs from the documented specification

2.64% of loans (845) have a production PD that differs from the PD recomputed from the specification (largest difference 0.023). 99% of the mismatches occur when current_ratio is missing: the production system imputes a different value from the documented one.

Recommendation. Align the production imputation with the documentation (or document the real rule) and test the implementation on a representative sample.

Area: Replication · Evidence: tables/implementation_diagnosis.csv · Owner: Model owner · Due: 2027-02-09 · Status: Open

F-05 Medium Reported calibration does not match the data

The back-test memo reports an observed / predicted default ratio of 1.00; recalculated on the validation sample it is 1.25.

Recommendation. Correct the back-test memo and recalibrate the PD (see the calibration finding).

Area: Replication · Evidence: tables/reported_vs_recalculated.csv · Owner: Model owner · Due: 2027-02-09 · Status: Open

F-06 Medium Model under- or over-predicts defaults

On the validation sample the observed default rate is 1.25x the mean predicted PD (7.05% against 5.66%).

Recommendation. Recalibrate the PD (intercept shift on recent vintages) and put a compensating margin in provisioning or pricing until done.

Area: Performance · Evidence: tables/calibration_validation.csv · Owner: Model owner · Due: 2027-02-09 · Status: Open

F-07 Medium Critical documentation items missing or partial

Critical items not met: Time of availability of every input (at decision / later) is documented; Assumptions and limitations of the model are stated; A monitoring plan with metrics and thresholds exists. Partly met: none.

Recommendation. Provide the missing critical documentation; without it the model cannot be re-implemented, monitored or defended.

Area: Documentation · Evidence: tables/documentation_review.csv · Owner: Model owner · Due: 2027-02-09 · Status: Open

F-08 Low An input's distribution has shifted

leverage has a CSI of 0.130 in 2023Q2.

Recommendation. Check data feeds and the applicant mix; assess the effect on the PD.

Area: Stability · Evidence: tables/csi.csv · Owner: Model owner · Due: 2027-05-10 · Status: Open

F-09 Low The model is weak for a segment

Gini for business_age = under 2 is 0.425, 74% of the overall 0.576.

Recommendation. Assess whether the model is fit for this segment; consider a separate model, overlay or manual review.

Area: Segments · Evidence: tables/segments.csv · Owner: Model owner · Due: 2027-05-10 · Status: Open

F-10 Low Documentation is incomplete

The documentation meets 74% of the validation checklist (4 items not met, 4 partly met).

Recommendation. Complete the missing and partial items of the checklist and re-submit the documentation.

Area: Documentation · Evidence: tables/documentation_review.csv · Owner: Model owner · Due: 2027-05-10 · Status: Open

F-11 Low Model used beyond its documented scope

Loans above 5m appear in the portfolio scored by the model although the development report excludes them (it states a maximum of 5m).

Recommendation. Either extend the model's documented scope with evidence or route these loans to a different process.

Area: Governance · Evidence: Interview with the credit risk team; loan_amount distribution in observations.csv · Owner: Model owner · Due: 2027-05-10 · Status: Open

Limitations and assumptions

- The validation relies on the data, specification and documents supplied by the lender; it does not audit the lender's source systems.
- Outcome analysis covers loans whose 12-month outcome window had elapsed at 2024-06-30; recent loans are not assessed.
- The production model is tested as delivered (production PD); the implementation check covers the PD, not the surrounding IT controls.
- Challenger models show what the data can support; they are not proposed replacements.
- Thresholds are those agreed at kickoff; severity ratings are judgements supported by the evidence in the tables.
- Findings from interviews and documents are listed as manual findings; they depend on the information provided.
- This demonstration uses synthetic data; its numbers have no meaning for any real lender.

Appendix

A.1 Configuration used

Samples	development to 2021-12-31; validation 2022-01-01 to 2022-12-31; monitoring from 2023-01-01
Outcome	90+ days past due within 12 months of origination; window 12 months; data as of 2024-06-30
Severity scale	High: not relied on until fixed (30 days); Medium: fix with compensating control (90 days); Low: improvement (180 days); Observation: no action
Stability thresholds	PSI / CSI: below 0.10 stable, 0.10 to 0.25 watch, above 0.25 breach

Source: config_used.yaml in the run folder

A.2 Data cleaning log

step	action	rows_affected	key
observations	drop exact duplicate rows	1	
observations	category spelling harmonised (case / spaces)	65	industry
observations	impossible values set to missing in the working copy (outside [0, 150])	6	years_in_business
observations	impossible values set to missing in the working copy (outside [1, None])	2	annual_revenue

Source: tables/cleaning_log.csv

Data quality

model_spec · 11 rows · PASSED (0 errors, 0 warnings)

Check	Severity	Column	Count	Message
missing_values	info	source	1	1 missing (9.09%)
missing_values	info	cap	6	6 missing (54.55%)
missing_values	info	fill	3	3 missing (27.27%)
missing_values	info	levels	9	9 missing (81.82%)

Source: data_quality/dq_model_spec.json

observations · 32052 rows · PASSED (0 errors, 4 warnings)

Check	Severity	Column	Count	Message
missing_values	info	current_ratio	837	837 missing (2.61%)
missing_values	info	interest_cover	672	672 missing (2.10%)
missing_values	info	relationship_years	465	465 missing (1.45%)
duplicate_rows	warning		1	1 fully duplicated rows
duplicate_keys	warning		2	2 rows share a key ['loan_id'] (0 with conflicting values)
out_of_range	warning	years_in_business	6	6 values outside the business range [0, 150]
out_of_range	warning	annual_revenue	2	2 values outside the business range [1, None]

Source: data_quality/dq_observations.json

observations_clean · 32051 rows · PASSED (0 errors, 0 warnings)

Check	Severity	Column	Count	Message
missing_values	info	years_in_business	6	6 missing (0.02%)
missing_values	info	annual_revenue	2	2 missing (0.01%)
missing_values	info	current_ratio	837	837 missing (2.61%)
missing_values	info	interest_cover	672	672 missing (2.10%)
missing_values	info	relationship_years	465	465 missing (1.45%)

Source: data_quality/dq_observations_clean.json

reported_metrics · 7 rows · PASSED (0 errors, 0 warnings)

No data-quality issues detected.

Audit trail

Run ID	`mv_20261111T142054Z_2fe5dd`
Run started / finished (UTC)	2026-11-11T14:20:54+00:00
Analyst	info@proquantanalytics.com
Code version (git)	`3a6bb15275c5 (uncommitted changes) · core 0.1.0`
Config SHA-256	`fd4156968f530c6c`
Input: observations	`32052 rows · sha256 d6f2b749f25ecb63`
Input: model_spec	`11 rows · sha256 5c643980e2193c3d`
Input: reported_metrics	`7 rows · sha256 1a26e8a36ad177ee`
Environment	`python 3.13.5 · pandas 2.2.3 · numpy 2.1.3`

Source: audit.json in the run folder

This report was prepared by ProQuant Analytics for Demo Client under the terms of the engagement agreement, from data and documents provided by the model owner, using the methodology described herein. It expresses an independent professional opinion on the model's fitness for the stated use at the date of the report and does not constitute legal, regulatory or credit advice. Statistical models describe past behaviour and can be wrong in new conditions.